Report

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Summary

General Tasks Status

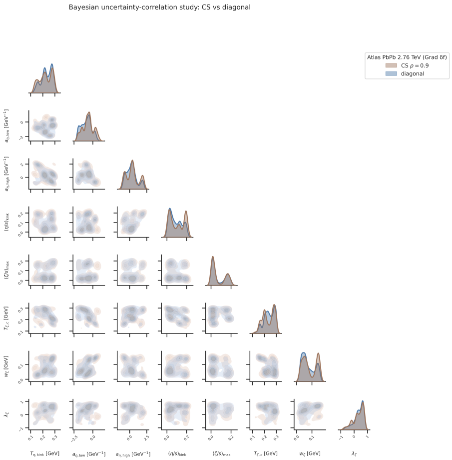
Task	**Status**	
2026-06 RETINHA presentation	◑	Link
2026-06 Scientific report	◑	Link

Task	**Status**	
Journal Club - Automations	◑	

Correlated Uncertainty Study Tasks Status

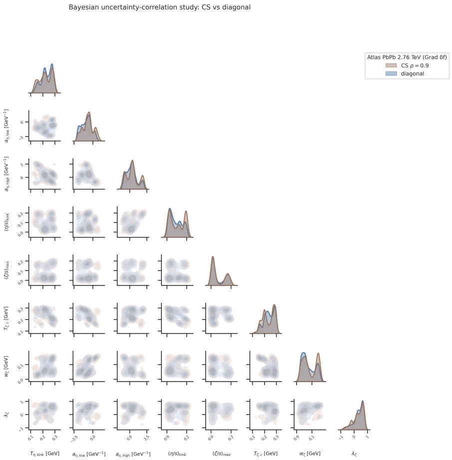
Task	Status	Link
Alice ins1222333 Studies		
↪ Mcmc Params	✓	Plots
↪ RDF Model	✓	Plots
↪ CS Model	✓	Plots
↪ None Uncertainty Correlation	◔	Could not initialize ptemcee walkers inside posterior support after 1000 attempts. Try a different seed or relax the initial ranges.
Atlas ins1759506 studies		
↪ Uncertainty correlation	◔	
RHIC Au-Au/d-Au Studies JETSCAPE		

Baseline vs CS correlation model



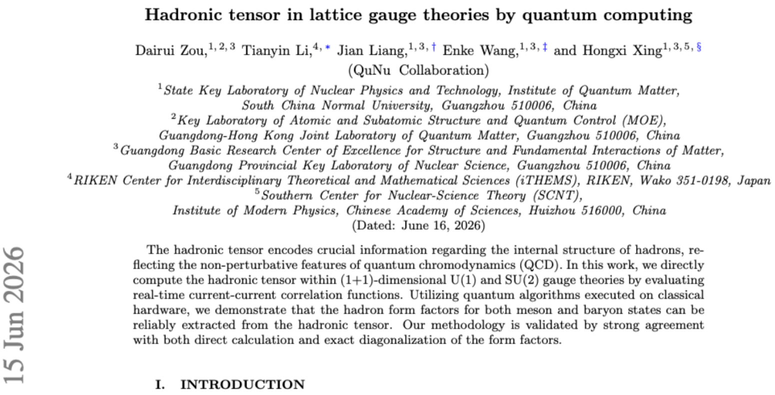
Figure

Baseline vs RDS correlation model



Figure

Selected Article



Hadronic tensor in lattice gauge theories by quantum computing

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The hadronic tensor encodes crucial information regarding the internal structure of hadrons, reflecting the non-perturbative features of quantum chromodynamics (QCD). In this work, we directly compute the hadronic tensor within (1+1)-dimensional U(1) and SU(2) gauge theories by evaluating real-time current-current correlation functions. Utilizing quantum algorithms executed on classical hardware, we demonstrate that the hadron form factors for both meson and baryon states can be reliably extracted from the hadronic tensor. Our methodology is validated by strong agreement with both direct calculation and exact diagonalization of the form factors.

I. INTRODUCTION

Figure: Lattice Quantum Chromo Dynamics (QCD) calculations on Quantum Computers